

DSBDA

Data Science & Big Data Analytics

Complete Viva Notes | Definitions | Comparisons | Applications | Adv & Disadv

Subject	DSBDA — Data Science & Big Data Analytics
Units	Big Data · Hadoop/MapReduce · Data Science · ML · NoSQL · Visualization · Statistics · Cloud
New	Pandas vs NumPy · Matplotlib vs Seaborn · Plot Types · Supervised vs Unsupervised · Batch vs Stream · SQL vs NoSQL

UNIT 1 — Big Data Overview

Big Data: Extremely large datasets that cannot be processed using traditional tools. Characterised by 5 Vs: Volume, Velocity, Variety, Veracity, Value.

Volume: Massive scale of data — from terabytes to zettabytes — generated every second from sensors, social media, transactions.

Velocity: Speed at which new data is generated and must be processed. Real-time processing is critical for applications like stock trading.

Variety: Multiple data types: Structured (RDBMS), Semi-structured (JSON, XML), Unstructured (images, video, text, audio).

Veracity: Uncertainty and trustworthiness of data. Incomplete or noisy data reduces analytics quality.

Value: The business usefulness extracted from processed data. Raw big data has little worth until converted into actionable insights.

Structured Data: Data organised in a fixed schema with rows and columns. Example: MySQL tables, Excel spreadsheets.

Unstructured Data: Data with no predefined format. Example: emails, social media posts, videos, audio files, PDFs.

Semi-Structured Data: Has some organisational properties but no strict schema. Example: JSON, XML, CSV files.

Data Lake: Centralised repository storing all structured and unstructured data at any scale in raw format until needed for analysis.

Data Warehouse: Stores current and historical, cleaned, structured data from multiple sources optimised for reporting and OLAP queries.

Metadata: Data about data. Describes the content, quality, condition, and other characteristics of data (e.g., file size, creation date, schema).

Applications of Big Data

Application Area	Description
Healthcare	Patient records analysis, disease prediction, drug discovery, genomics, personalised medicine.
Finance & Banking	Fraud detection, risk management, algorithmic trading, credit scoring, KYC.
Retail & E-commerce	Customer behaviour analysis, recommendation engines, demand forecasting, inventory management.
Telecom	Network optimisation, customer churn prediction, call data records (CDR) analysis.
Government	Smart cities, crime analytics, election data, census analysis, tax fraud detection.
Manufacturing	Predictive maintenance, supply chain optimisation, defect detection, quality control.
Social Media	Sentiment analysis, trend detection, ad targeting (Facebook, Twitter, Instagram).
Transport & Logistics	Route optimisation, traffic prediction, autonomous vehicles, fleet management.

Advantages vs Disadvantages of Big Data

✓ Advantages	✗ Disadvantages
• Faster & better decision making	• High storage and processing cost
• Fraud & anomaly detection	• Data privacy & security risks

• Improved customer experience	• Data quality and accuracy issues
• Cost reduction via optimisation	• Requires specialised skills
• Real-time business insights	• Complex infrastructure management
• Competitive advantage	• Risk of data breaches
• Enables personalization at scale	• Regulatory compliance challenges

UNIT 2 — Hadoop Ecosystem & MapReduce

Hadoop: Open-source Java-based framework for distributed storage and processing of large datasets across commodity hardware clusters.

HDFS: Hadoop Distributed File System — splits files into blocks (default 128 MB), replicates them (default 3x) across DataNodes for fault tolerance.

NameNode: Master node of HDFS storing metadata (file names, block locations, directory structure). Single point of coordination.

DataNode: Slave nodes that store actual data blocks. Send periodic heartbeats to NameNode to confirm health.

MapReduce: Programming model for parallel processing. Two phases: Map (splits input into key-value pairs) and Reduce (aggregates key-value pairs to output).

Map Phase: Input data is split and processed in parallel. Each mapper emits intermediate key-value pairs.

Shuffle & Sort: The framework automatically groups all values for the same key together before passing to reducers.

Reduce Phase: Reducers process grouped key-value pairs and produce the final output written to HDFS.

YARN: Yet Another Resource Negotiator — separates resource management from data processing; manages CPU and memory allocation across the cluster.

Hive: Data warehouse on Hadoop; provides HiveQL (SQL-like) to query data stored in HDFS. Converts SQL queries to MapReduce/Tez/Spark jobs.

Pig: High-level scripting platform using Pig Latin language for data transformation on Hadoop; easier than raw MapReduce.

HBase: Distributed, column-oriented NoSQL database on HDFS for random, real-time read/write access to billions of rows.

Sqoop: Bulk data transfer tool between Hadoop and RDBMS (MySQL, Oracle). Supports import and export.

Flume: Distributed service for collecting, aggregating, and moving large volumes of log/event data into HDFS.

Zookeeper: Centralised coordination service for distributed applications — configuration management, naming, synchronisation.

Apache Spark: Fast, in-memory cluster computing engine. Processes data up to 100x faster than MapReduce. Supports batch, streaming, ML, and graph processing.

MapReduce vs Apache Spark — Detailed Comparison

Feature	MapReduce	Apache Spark
Processing Model	Batch processing only	Batch + Streaming + Interactive
Speed	Slow — disk-based I/O every step	100x faster — in-memory processing
Data Storage	Reads/writes HDFS at every step	Stores data in-memory (RDD/DataFrame)
Ease of Use	Complex Java API	Python/Scala/R/Java support, easy API
Fault Tolerance	Re-runs failed tasks from HDFS	RDD lineage-based re-computation
Real-time Processing	Not supported	Spark Streaming / Structured Streaming
Machine Learning	Mahout (limited)	MLlib (rich ML library built-in)
Graph Processing	Not supported natively	GraphX library
Iterative Algorithms	Very slow (repeated disk I/O)	Fast — data cached in memory

Resource Manager	YARN / Standalone	YARN / Mesos / Kubernetes / Standalone
Best For	Simple batch ETL jobs	Complex analytics, ML, real-time

Advantages vs Disadvantages — Hadoop

✓ Advantages	✗ Disadvantages
• Handles petabyte-scale data	• Not suitable for real-time processing
• Cost-effective (commodity hardware)	• High latency for small files
• Fault-tolerant (data replication x3)	• Java expertise required
• Highly scalable (add nodes)	• No native OLTP support
• Supports structured + unstructured data	• Security features are basic
• Strong open-source ecosystem	• Complex setup and debugging

Advantages vs Disadvantages — MapReduce

✓ Advantages	✗ Disadvantages
• Massive parallel processing	• Disk-based — high latency
• Works on commodity hardware	• Not good for iterative algorithms
• Automatic fault tolerance	• High programming overhead
• Scales to thousands of nodes	• No real-time support

UNIT 3 — Data Science & Analytics

Data Science: Interdisciplinary field combining statistics, mathematics, computer science, and domain knowledge to extract insights from data.

Data Scientist: Professional who uses ML, statistics, and programming to extract knowledge from structured and unstructured data.

Data Analytics: Process of examining datasets to draw conclusions. Types: Descriptive, Diagnostic, Predictive, Prescriptive.

Descriptive Analytics: Summarises past data — what happened? Uses averages, percentages, dashboards.

Diagnostic Analytics: Analyses why something happened using drill-down and data discovery techniques.

Predictive Analytics: Forecasts future events using ML models and historical patterns (e.g., churn prediction).

Prescriptive Analytics: Recommends best actions given predictions using optimisation and simulation techniques.

CRISP-DM: Cross-Industry Standard Process for Data Mining: Business Understanding → Data Understanding → Data Preparation → Modelling → Evaluation → Deployment.

ETL: Extract, Transform, Load — process of extracting data from sources, transforming to required format, loading into a data warehouse.

Feature Engineering: Creating or selecting input variables from raw data to improve model performance.

Overfitting: Model memorises training data including noise; performs well on training but poorly on unseen data. Solution: regularisation, cross-validation.

Underfitting: Model too simple to capture patterns; performs poorly on training and test data. Solution: more features, complex models.

Cross-Validation: Splits data into k folds; trains/tests on different combinations to reliably estimate model performance.

Train-Test Split: Divides dataset into training set (usually 70-80%) and testing set (20-30%) to evaluate model generalisation.

Data Preprocessing: Cleaning raw data: handling missing values, encoding categorical variables, normalisation/standardisation, removing outliers.

Normalisation: Scales data to a range [0, 1] using min-max formula: $(x - \min) / (\max - \min)$. Sensitive to outliers.

Standardisation: Scales data to have mean=0 and std=1 using Z-score: $(x - \text{mean}) / \text{std}$. Robust to outliers.

4 Types of Analytics — Comparison

Type	Question Answered	Techniques	Example
Descriptive	What happened?	Aggregation, summaries, reports, dashboards	Monthly revenue report
Diagnostic	Why did it happen?	Drill-down, data mining, correlation analysis	Why did sales drop in Q3?
Predictive	What will happen?	ML models, regression, forecasting	Customer churn prediction
Prescriptive	What should we do?	Optimization, simulation, decision trees	Recommend best marketing campaign

UNIT 4 — Machine Learning

Machine Learning: Branch of AI enabling computers to learn from data without explicit programming. Improves through experience.

Supervised Learning: Trains on labelled data (input+output pairs). Goal: learn input→output mapping. Types: Classification & Regression.

Unsupervised Learning: Finds hidden patterns in unlabelled data. Types: Clustering, Dimensionality Reduction, Association.

Semi-Supervised Learning: Uses small amount of labelled data + large amount of unlabelled data. Reduces annotation cost.

Reinforcement Learning: Agent learns via trial-and-error by receiving rewards/penalties from environment interactions. Used in games, robotics.

Classification: Predicts discrete class labels. Output is a category. Example: Spam/Not Spam, Disease/No Disease.

Regression: Predicts continuous numeric values. Output is a number. Example: House price prediction, Stock price.

Clustering: Groups similar data points without labels. K-Means, DBSCAN, Hierarchical Clustering.

Linear Regression: Models relationship between variables with a straight line $y = mx + c$. Minimises MSE (Mean Squared Error).

Logistic Regression: Binary classification using sigmoid function. Outputs probability between 0 and 1.

Decision Tree: Tree structure where nodes=feature tests, branches=outcomes, leaves=class labels. Prone to overfitting.

Random Forest: Ensemble of multiple decision trees. Reduces overfitting via bagging and feature randomness.

SVM: Finds optimal separating hyperplane that maximises margin between classes. Effective in high-dimensional spaces.

KNN: Classifies based on majority class of K nearest neighbours. Lazy learner — no training phase.

K-Means: Unsupervised clustering: partitions n points into K clusters by minimising intra-cluster variance (distance to centroid).

Naive Bayes: Probabilistic classifier using Bayes' theorem assuming feature independence. Very fast, good for NLP.

Neural Network: Layers of interconnected neurons. Input layer → Hidden layers → Output layer. Learns weights via backpropagation.

Deep Learning: Multi-layer neural networks that automatically learn hierarchical features from raw data (images, audio, text).

Confusion Matrix: Evaluation table: TP (correct positives), TN (correct negatives), FP (false alarms), FN (missed positives).

Precision: $TP/(TP+FP)$. Of all predicted positives, how many are actually positive. High when false alarms are costly.

Recall: $TP/(TP+FN)$. Of all actual positives, how many were correctly found. High when missing positives is costly.

F1-Score: $2 \cdot (\text{Precision} \cdot \text{Recall}) / (\text{Precision} + \text{Recall})$. Harmonic mean — balances precision and recall.

Accuracy: $(TP+TN)/(TP+TN+FP+FN)$. Overall correctness. Misleading for imbalanced datasets.

ROC-AUC: Plots True Positive Rate vs False Positive Rate. AUC=1 is perfect. AUC=0.5 is random guessing.

Gradient Descent: Optimisation algorithm that iteratively updates model weights to minimise the loss function.

Regularisation: Technique to prevent overfitting: L1 (Lasso) adds sum of absolute weights; L2 (Ridge) adds sum of squared weights to loss.

Supervised vs Unsupervised vs Reinforcement Learning

Feature	Supervised	Unsupervised	Reinforcement
Data Required	Labelled data (input+output)	Unlabelled data (input only)	No dataset — learns from environment
Goal	Learn input-output mapping	Find hidden patterns/structure	Maximise cumulative reward
Output	Predicted label or value	Clusters or reduced dimensions	Policy (sequence of actions)
Algorithms	Linear Reg, SVM, Random Forest	K-Means, PCA, DBSCAN	Q-Learning, DQN, PPO
Feedback	Direct (correct labels known)	No direct feedback	Reward/penalty from environment
Examples	Spam detection, price prediction	Customer segmentation, anomaly	Chess AI, robot navigation
Difficulty	Medium	Hard (no ground truth)	Very hard (delayed rewards)

Classification vs Regression — Comparison

Feature	Classification	Regression
Output Type	Discrete class labels	Continuous numeric values
Goal	Predict category/class	Predict a quantity
Algorithms	Logistic Reg, SVM, Decision Tree, KNN, Random Forest	Linear Reg, Ridge, Lasso, SVR, Neural Net
Evaluation	Accuracy, Precision, Recall, F1, ROC-AUC	MSE, RMSE, MAE, R-squared
Examples	Spam/Not-Spam, Disease Yes/No, Digit recognition	House price, Temperature, Stock price
Loss Function	Cross-Entropy / Log Loss	MSE / MAE / RMSE

Advantages vs Disadvantages — Machine Learning

✓ Advantages	✗ Disadvantages
• Learns automatically from data	• Needs large labelled data
• Handles complex high-dimensional data	• Black-box — low interpretability
• Scales with more data	• Computationally expensive
• Discovers hidden patterns	• Sensitive to data quality
• Applicable across all domains	• Biased if training data is biased

UNIT 5 — NoSQL Databases

NoSQL: Not Only SQL — non-relational databases designed for horizontal scalability, flexible schemas, and handling large volumes of varied data.

Document Store: Stores data as JSON/BSON documents. Flexible schema per document. Example: MongoDB, CouchDB.

Key-Value Store: Simplest model — data stored as key-value pairs. Extremely fast for simple lookups. Example: Redis, DynamoDB, Riak.

Column-Family Store: Stores data by columns instead of rows. Optimised for wide-column queries. Example: Cassandra, HBase.

Graph Database: Nodes store entities; edges store relationships. Ideal for social networks, fraud detection. Example: Neo4j, Amazon Neptune.

MongoDB: Document-oriented NoSQL using BSON format. Supports dynamic schemas, replication, horizontal sharding, geospatial queries.

Cassandra: Distributed wide-column store. Masterless peer-to-peer architecture. Designed for zero downtime, linear scalability.

Redis: In-memory key-value store. Sub-millisecond latency. Used as cache, session store, message broker, leaderboard.

CAP Theorem: A distributed system can guarantee only 2 of 3: Consistency (all nodes see same data), Availability (system always responds), Partition Tolerance (works despite network failures).

Sharding: Horizontally partitioning data across multiple nodes. Each shard holds a subset of total data. Improves read/write throughput.

Replication: Copying data across multiple nodes. Ensures high availability and fault tolerance. Types: Master-Slave, Peer-to-Peer.

ACID: Atomicity, Consistency, Isolation, Durability — ensures reliable transactions in RDBMS.

BASE: Basically Available, Soft-state, Eventually Consistent — NoSQL alternative to ACID; prioritises availability over immediate consistency.

Eventual Consistency: All nodes will eventually converge to the same value, but may be temporarily inconsistent during propagation.

NoSQL vs RDBMS — Detailed Comparison

Feature	RDBMS	NoSQL
Schema	Fixed, predefined schema	Flexible / schema-less / dynamic
Scalability	Vertical (scale up — better hardware)	Horizontal (scale out — add nodes)
Transactions	ACID compliant	BASE compliant (eventual consistency)
Query Language	Standard SQL	Varies: MQL (Mongo), CQL (Cassandra)
Data Model	Tables with rows and columns	Document, Key-Value, Column, Graph
Joins	Supports complex JOINS natively	No JOIN support; denormalized data
Best For	Structured relational data, transactions	Big data, unstructured, fast scaling
Performance	Slows with very large scale	High performance at petabyte scale
ACID vs BASE	Strongly ACID	Eventually consistent (BASE)

Examples	MySQL, PostgreSQL, Oracle, SQL Server	MongoDB, Cassandra, Redis, HBase
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4 NoSQL Types — Side-by-Side Comparison

Feature	Key-Value	Document	Column-Family	Graph
Data Model	key → value	JSON/BSON documents	Column families	Nodes & Edges
Schema	Schema-less	Flexible per doc	Fixed column fam.	Schema-less
Query	By key only	Rich queries	Column-range scans	Graph traversal
Best For	Caching, sessions	CMS, user profiles	Time-series, IoT	Social networks
Examples	Redis, DynamoDB	MongoDB, CouchDB	Cassandra, HBase	Neo4j, Neptune
Scalability	Very High	High	Very High	Moderate

Advantages vs Disadvantages — NoSQL

✓ Advantages	✗ Disadvantages
• Flexible schema design	• Limited ACID transaction support
• Horizontal scalability	• No standardised query language
• High performance at scale	• Eventual consistency issues
• Handles all data types	• Weaker security features
• High availability & fault tolerance	• Less mature ecosystem than RDBMS

UNIT 6 — Python Libraries for Data Science

NumPy: Numerical Python — fundamental package for scientific computing. Provides N-dimensional arrays (ndarray), mathematical operations, linear algebra, Fourier transforms.

Pandas: Python Data Analysis Library — provides DataFrame and Series data structures. Used for data manipulation, cleaning, merging, grouping, and analysis.

Matplotlib: Python 2D plotting library. Foundation of most Python visualisation tools. Low-level, highly customisable, supports line, bar, scatter, histogram, pie charts.

Seaborn: Built on Matplotlib; provides high-level interface for statistical graphics. Beautiful default styles, built-in statistical visualisations.

Scikit-learn: ML library for Python. Classification, regression, clustering, dimensionality reduction, model selection, preprocessing tools.

TensorFlow: Open-source ML framework by Google. Used for deep learning, neural networks, large-scale ML pipelines.

Keras: High-level neural network API running on top of TensorFlow. User-friendly for building deep learning models.

PyTorch: Open-source deep learning framework by Facebook/Meta. Dynamic computation graph, popular in research.

SciPy: Scientific computing library extending NumPy. Statistics, signal processing, optimization, linear algebra.

Plotly: Interactive visualisation library for Python. Creates web-based interactive charts, dashboards.

NumPy vs Pandas — Detailed Comparison

Feature	NumPy	Pandas
Primary Use	Numerical computation & arrays	Data manipulation & analysis
Data Structure	ndarray (N-dimensional array)	DataFrame (2D) & Series (1D)
Data Types	Homogeneous (same dtype)	Heterogeneous (mixed dtypes per column)
Index	Integer-based (0, 1, 2...)	Labelled index (integers, strings, dates)
Missing Values	No native NaN handling	Native NaN support with fillna/dropna
Operations	Element-wise math, broadcasting	Group-by, merge, pivot, reshape, apply
Speed	Faster for pure math operations	Slower but richer data operations
File I/O	Limited (npy, txt)	Rich: CSV, Excel, JSON, SQL, HDF5, Parquet
Column Access	df[:, 0] — index based	df['col'] — name based
Use Case	Linear algebra, signal processing, ML backends	Data wrangling, EDA, preprocessing
Memory	Less overhead (typed arrays)	More overhead (flexible dtypes)
Syntax Example	np.mean(arr), arr.reshape(3,4)	df.groupby('col').mean(), df.merge(df2)

Matplotlib vs Seaborn — Detailed Comparison

Feature	Matplotlib	Seaborn
Level	Low-level — fine-grained control	High-level — built on Matplotlib
Syntax	Verbose (many lines of code)	Concise (1-2 lines for complex plots)

Default Aesthetics	Basic, needs manual styling	Beautiful defaults, statistical themes
Statistical Plots	Manual implementation required	Built-in: boxplot, violin, pairplot, heatmap
DataFrame Support	Needs manual data extraction	Native Pandas DataFrame integration
Color Palettes	Basic defaults	Rich palettes: deep, muted, pastel, dark
Complex Plots	Full control, highly customisable	Easier for statistical visualisations
3D Plotting	Supported (mpl_toolkits.mplot3d)	Not supported natively
Interactivity	Static (use with mpld3 for interact)	Static (use Plotly for interactivity)
Best For	Custom plots, publication-quality figs	Quick statistical analysis, EDA
Common Plots	Line, bar, scatter, histogram, pie	heatmap, pairplot, violin, jointplot, Implot
Import	import matplotlib.pyplot as plt	import seaborn as sns

All Major Plot Types — When to Use What

Plot Type	Use When	Variables	Library	Example Use Case
Line Plot	Showing trend over time	1 continuous X + 1 continuous Y	Matplotlib / Seaborn	Stock price over months
Bar Chart	Comparing categories	1 categorical + 1 numeric	Matplotlib / Seaborn	Sales by region
Histogram	Showing distribution of 1 variable	1 continuous variable	Matplotlib / Seaborn	Age distribution
Scatter Plot	Showing relationship between 2 variables	2 continuous variables	Matplotlib / Seaborn	Height vs Weight
Box Plot	Showing spread + outliers	1 categorical + 1 numeric	Seaborn / Matplotlib	Salary by department
Violin Plot	Distribution + box plot combined	1 categorical + 1 numeric	Seaborn	Score distribution by class
Heatmap	Showing correlation matrix / 2D data	2D matrix / correlation table	Seaborn	Feature correlation matrix
Pie Chart	Showing proportions (few categories)	1 categorical with proportions	Matplotlib	Market share breakdown
Pair Plot	Relationships between all feature pairs	Multiple numeric variables	Seaborn	Iris dataset EDA
KDE Plot	Smooth probability density estimate	1-2 continuous variables	Seaborn	Data distribution shape
Joint Plot	Scatter + distribution together	2 continuous variables	Seaborn	Bivariate analysis
Count Plot	Frequency of each category	1 categorical variable	Seaborn	Gender distribution in survey

UNIT 7 — Statistics & Probability for Data Science

Mean: Arithmetic average = sum of all values / count. Sensitive to outliers.

Median: Middle value of sorted data. Robust to outliers. Use when distribution is skewed.

Mode: Most frequently occurring value. Used for categorical data.

Standard Deviation: Square root of variance. Measures spread around mean. Low SD = data clustered near mean.

Variance: Average squared deviation from mean. $\text{Var} = \text{SD}^2$. Measures data dispersion.

Normal Distribution: Symmetric bell-shaped distribution. Mean = Median = Mode. 68-95-99.7 rule: 1/2/3 SDs cover 68/95/99.7% of data.

Skewness: Asymmetry of distribution. Right skew: tail to right, mean > median. Left skew: tail to left, mean < median.

Kurtosis: Measure of tail heaviness. High kurtosis = more extreme outliers (heavy tails).

Pearson Correlation: Linear relationship between two variables. Range: -1 to +1. +1 = perfect positive, -1 = perfect negative, 0 = no linear relationship.

Spearman Correlation: Rank-based correlation. Used when data is not normally distributed or has outliers.

Covariance: Measures how two variables change together. Positive: both increase together; Negative: one increases as other decreases.

Hypothesis Testing: Statistical test to verify if a claim about population is supported. H_0 (null) vs H_1 (alternate hypothesis).

p-value: Probability of observing results as extreme as actual, assuming H_0 is true. $p < 0.05 \rightarrow$ reject H_0 (5% significance level).

Type I Error: False Positive — rejecting H_0 when it is actually true. Significance level alpha controls this.

Type II Error: False Negative — failing to reject H_0 when it is actually false. Power of test controls this.

Confidence Interval: Range likely to contain the true population parameter with specified confidence (e.g., 95% CI).

Central Limit Theorem: Sampling distribution of sample mean approaches normal distribution as n increases, regardless of population distribution.

Bayes Theorem: $P(A|B) = P(B|A) \cdot P(A) / P(B)$. Updates probability of a hypothesis based on new evidence.

Chi-Square Test: Tests independence between two categorical variables using expected vs observed frequencies.

ANOVA: Analysis of Variance — tests if means of 3+ groups are significantly different.

Mean vs Median vs Mode — When to Use

Measure	Formula	When to Use	Affected by Outliers?	Data Type
Mean	Sum/Count	Symmetric, normal distribution	Yes — highly	Numeric
Median	Middle value	Skewed data, outliers present	No — robust	Numeric
Mode	Most frequent	Categorical data, finding peaks	No	Numeric or Categorical

Correlation vs Covariance — Comparison

Feature	Correlation	Covariance
Definition	Normalised measure of linear relationship	Measure of joint variability
Range	-1 to +1	Unbounded (-inf to +inf)
Interpretation	Clear: +1 perfect pos, -1 perfect neg, 0 none	Difficult: only sign indicates direction
Units	Dimensionless (unit-free)	Product of units of two variables
Comparability	Comparable across different data	Not directly comparable across datasets
Formula	$\text{Cov}(X,Y) / (\text{SD}_X * \text{SD}_Y)$	$E[(X-\mu_X)(Y-\mu_Y)]$

UNIT 8 — Cloud Computing & Data Streaming

Cloud Computing: Delivery of computing services (servers, storage, databases, ML, analytics) over the internet on a pay-as-you-go model.

IaaS: Infrastructure as a Service — virtualised computing infrastructure. Example: AWS EC2, Google Compute Engine, Azure VMs.

PaaS: Platform as a Service — platform for building/deploying apps without managing underlying infra. Example: Google App Engine, Heroku.

SaaS: Software as a Service — cloud-hosted software accessed via browser. Example: Gmail, Salesforce, Dropbox, Office 365.

AWS: Amazon Web Services — leading cloud platform. Key services: S3 (storage), EC2 (compute), EMR (Hadoop/Spark), Redshift (DW), SageMaker (ML).

Apache Kafka: Distributed, high-throughput message streaming platform. Producers write to topics; Consumers read from topics. Used in LinkedIn, Uber, Netflix.

Apache Storm: Real-time distributed computation system. Guarantees message processing. Low-latency stream processing.

Apache Flink: Stateful stream processing framework for unbounded (streaming) and bounded (batch) data. True streaming, not micro-batching.

Streaming Data: Continuous data generated from IoT sensors, clickstreams, social media, logs — processed in real time.

Batch Processing: Collects and processes large data volumes at scheduled intervals (not real time). Example: overnight ETL jobs.

Lambda Architecture: Combines batch layer (Hadoop) + speed layer (Spark Streaming/Storm) + serving layer. Handles both historical and real-time data.

Kappa Architecture: Simplified Lambda — everything treated as a stream. Single processing path. Easier to maintain.

IoT: Network of physical devices with sensors/actuators exchanging data over internet. Generates massive real-time data streams.

Pub-Sub Model: Publisher-Subscriber messaging pattern — publishers send to topics; subscribers receive from topics. Kafka implements this model.

Batch Processing vs Stream Processing — Comparison

Feature	Batch Processing	Stream Processing
Data Type	Bounded (finite dataset)	Unbounded (continuous, infinite)
Processing Time	Hours to days (scheduled jobs)	Milliseconds to seconds (real-time)
Latency	High latency (not real-time)	Very low latency
Throughput	High (processes large volumes at once)	Lower per query, but continuous
Use Cases	Monthly reports, historical analytics	Fraud detection, live monitoring
Tools	Hadoop MapReduce, Apache Hive, Spark	Kafka, Spark Streaming, Flink, Storm
Storage	Stores all data before processing	Processes data on the fly
Fault Recovery	Restart failed job from checkpoint	State management + replay from Kafka
Cost	Lower infra cost	Higher infra cost for low-latency

Examples	End-of-day bank transactions, ETL	Real-time recommendations, IoT alerts
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IaaS vs PaaS vs SaaS — Comparison

Feature	IaaS	PaaS	SaaS
What is managed	VMs, storage, networking	Runtime, middleware, OS	Complete application
User manages	OS, runtime, apps, data	Apps, data only	Nothing (just use it)
Flexibility	Maximum flexibility	Medium flexibility	Minimal flexibility
Target Users	IT admins, DevOps	Developers	End users
Scaling	Manual or auto-scale	Auto-scale by platform	Fully managed by provider
Cost Model	Pay per resource used	Pay per usage	Subscription-based
Examples	AWS EC2, Azure VMs, GCE	Heroku, App Engine, Azure App Service	Gmail, Salesforce, Zoom

Advantages vs Disadvantages — Cloud Computing

✓ Advantages	✗ Disadvantages
• No upfront infra cost	• Data privacy & compliance concerns
• Elastic scalability on demand	• Ongoing subscription costs
• Managed services reduce overhead	• Vendor lock-in risk
• Global availability (multi-region)	• Requires internet connectivity
• Automatic updates & maintenance	• Latency issues for real-time apps
• Pay-as-you-go model	• Limited physical control

UNIT 9 — Data Visualization

Data Visualization: Graphical representation of data using charts, graphs, and maps to communicate insights visually.

Dashboard: Interactive display consolidating multiple visualisations, KPIs, and metrics on one screen.

Tableau: Drag-and-drop BI tool connecting to various data sources; creates interactive dashboards without coding.

Power BI: Microsoft BI tool; provides interactive visualisations and reports with DAX query language and strong Excel integration.

D3.js: Data-Driven Documents — JavaScript library for dynamic, interactive web-based visualisations using SVG/HTML/CSS.

Matplotlib: Python 2D plotting library. Foundation for Seaborn and Pandas plotting. Highly customisable.

Seaborn: Python statistical visualization on top of Matplotlib. Beautiful defaults, statistical plot types built-in.

Plotly: Interactive Python visualization library. Creates web-embedded charts and dashboards. Supports 3D, maps.

Bokeh: Python interactive visualization library targeting web browsers. Suitable for large/streaming datasets.

Heat Map: Color-gradient matrix showing magnitude of values. Used for correlation matrices and geographic density.

Box Plot: Shows 5-number summary: Min, Q1, Median, Q3, Max plus outliers. Reveals distribution and spread.

Scatter Plot: 2D plot of data points showing relationship/correlation between two numerical variables.

Histogram: Bar chart of frequency distribution showing how data is spread across bins/ranges.

Word Cloud: Visual text representation where word size reflects frequency or importance.

Choropleth Map: Geographic map where regions are shaded based on the value of a variable (e.g., GDP by country).

Tableau vs Power BI vs Matplotlib vs Seaborn

Feature	Tableau	Power BI	Matplotlib	Seaborn
Type	BI Tool	BI Tool	Python Library	Python Library
Coding	No code (drag-drop)	Minimal (DAX)	Requires Python	Requires Python
Interactivity	High	High	Low (static)	Low (static)
Cost	Expensive (paid)	Free + paid Pro	Free (open source)	Free (open source)
Integration	Many connectors	Microsoft ecosystem	Python ecosystem	Python ecosystem
Ease	Very easy	Easy	Steep learning curve	Easier than Mpl
Best For	Business dashboards	MS/Excel users	Custom plots	Statistical EDA
3D Charts	Limited	Limited	Supported	Not supported

Advantages vs Disadvantages — Data Visualization

✓ Advantages	✗ Disadvantages
• Simplifies complex data	• Misleading if poorly designed
• Faster pattern recognition	• Over-simplification risk
• Improves insight communication	• Requires design knowledge

• Helps decision making	• Interactive tools are expensive
• Identifies outliers quickly	• Too many visuals cause overload

UNIT 10 — Key Comparisons & Differences

Data Lake vs Data Warehouse vs Database

Feature	Database	Data Warehouse	Data Lake
Purpose	Operational OLTP	Analytical OLAP reporting	Store raw data of all types
Schema	Schema on write (fixed)	Schema on write (fixed)	Schema on read (flexible)
Data Type	Structured only	Structured, historical	All: structured/semi/unstructured
Processing	Real-time transactions	Batch analytical queries	Batch + streaming
Users	Application developers	Data analysts, BI teams	Data scientists, ML engineers
Query Speed	Very fast for transactions	Fast for analytical queries	Slower without optimization
Cost	Moderate	High (expensive hardware)	Low (commodity/cloud storage)
Examples	MySQL, Oracle, PostgreSQL	Redshift, Snowflake, Teradata	AWS S3, Azure Data Lake, HDFS

Hadoop vs Spark vs Flink — Three-Way Comparison

Feature	Hadoop	Spark	Flink
Processing	Batch only	Batch + micro-batch streaming	True streaming + batch
Speed	Slow (disk I/O)	100x faster (in-memory)	Very fast (pipelined streaming)
Latency	High (minutes-hours)	Medium (seconds-minutes)	Very low (milliseconds)
State Management	Limited	Limited in streaming	Advanced stateful operations
Fault Tolerance	HDFS replication	RDD lineage / checkpointing	Lightweight distributed snapshots
ML Support	Mahout	MLlib (rich)	FlinkML (basic)
API	Java MapReduce API	Python/Scala/Java/R	Java/Scala/Python
Best For	Large batch ETL	Interactive + iterative ML	Real-time event processing

Deep Learning vs Machine Learning vs AI — Hierarchy

Feature	AI (Artificial Intelligence)	Machine Learning	Deep Learning
Definition	Machines simulating human intelligence	Subset of AI; learns from data	Subset of ML; multi-layer neural nets
Scope	Broadest field	Narrower than AI	Narrowest, most specialised
Data Needs	Varies	Moderate labelled data	Huge amounts of data required
Feature Eng	Manual or rule-based	Manual feature engineering needed	Automatic feature learning
Hardware	General	CPU sufficient	GPU/TPU required
Explainability	Rule-based AI is explainable	Moderate explainability	Black-box — very low explainability
Applications	Chess, expert systems	Spam detection, recommendation	Image recognition, NLP, self-driving

Overfitting vs Underfitting — Comparison

Feature	Overfitting	Underfitting
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Definition	Model learns noise in training data	Model too simple to capture patterns
Training Acc.	Very High	Low
Test Acc.	Very Low	Low
Bias	Low bias	High bias
Variance	High variance	Low variance
Cause	Too complex model, too many features	Too simple model, few features
Solution	Regularization, Dropout, Cross-validation, Pruning	More features, complex model, more data
Analogy	Memorising exam answers — fails new questions	Not studying enough — fails exam

Precision vs Recall vs F1 — When to Use

Metric	Formula	High Value Means	Use When	Example
Precision	$TP/(TP+FP)$	Few false alarms	FP is costly	Spam filter — don't mark real emails as spam
Recall	$TP/(TP+FN)$	Few missed positives	FN is costly	Cancer diagnosis — don't miss a sick patient
F1-Score	$2*P*R/(P+R)$	Good balance of both	Class imbalance present	Fraud detection with imbalanced data
Accuracy	$(TP+TN)/All$	Overall correct %	Balanced classes	Image classification on balanced dataset

QUICK VIVA Q&A; — Must Know!

	A: Volume, Velocity, Variety, Veracity, Value.
	A: 128 MB (older Hadoop versions: 64 MB).
	A: NameNode = master; stores metadata. DataNode = slave; stores actual data.
	A: Spark is in-memory (100x faster); MapReduce is disk-based. Spark supports DAG.
	A: Model learns noise — high train accuracy, low test accuracy. Fix: regularisation.
	A: Consistency, Availability, Partition Tolerance. A distributed system can guarantee 2 out of 3.
	A: ACID (RDBMS): strict consistency. BASE (NoSQL): eventual consistency, availability, partition tolerance.
	A: Supervised: labelled data, predicts output. Unsupervised: unlabelled data, finds patterns.
	A: Table with TP, TN, FP, FN — evaluates classification model performance.
	A: NumPy: N-dim arrays, math ops. Pandas: DataFrame, data wrangling, analysis.
	A: Matplotlib: low-level, full control. Seaborn: high-level, beautiful defaults, statistical plots.
	A: Precision when FP is costly (spam filter). Recall when FN is costly (disease screening).
	A: Yet Another Resource Negotiator — separates resource management from application logic.
use?	A: Data Lake: raw data, schema on read. Data Warehouse: structured, processed data.
	A: Real-time data streaming. Pub-Sub messaging system. Used in LinkedIn, Uber, etc.
	A: Probability of observing results as extreme as actual, assuming H0 is true.
	A: Key-Value (Redis), Document (MongoDB), Column-Family (Cassandra), Graph (Neo4j).
	A: Unsupervised ML — partitions n data points into K clusters by minimising cost function.
	A: Batch: large data at scheduled intervals, high latency. Stream: continuous data, low latency.
	A: Sample mean distribution approaches normal distribution as sample size increases (Central Limit Theorem).
	A: Technique to prevent overfitting. L1 (Lasso): penalises absolute weights. L2 (Ridge): penalises squared weights.
	A: $P(A B) = \frac{P(B A) \cdot P(A)}{P(B)}$. Updates probability of hypothesis given new evidence.
	A: Combines batch layer + speed layer + serving layer to handle both historical & real-time data.
	A: Area Under ROC Curve. Measures classifier quality. AUC=1 perfect, AUC=0.5 random.
	A: Hive: SQL-like (HiveQL), best for structured data & DW queries. Pig: script-based, flexible.